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Gefitinib Alone Versus Gefitinib Plus Chemotherapy for Non–Small-Cell Lung Cancer With Mutated Epidermal Growth Factor Receptor: NEJ009 Study

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abstract

PURPOSE Epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor combined with cytotoxic chemotherapy is highly effective for the treatment of advanced non–small-cell lung cancer (NSCLC) with EGFR mutations; however, little is known about the efficacy and safety of this combination compared with that of standard therapy with EGFR- tyrosine kinase inhibitors alone.

METHODS We randomly assigned 345 patients with newly diagnosed metastatic NSCLC with EGFR mutations to gefitinib combined with carboplatin plus pemetrexed or gefitinib alone. Progression-free survival (PFS), PFS2, and overall survival (OS) were sequentially analyzed as primary end points according to a hierarchical sequential testing method. Secondary end points were objective response rate (ORR), safety, and quality of life.

RESULTS The combination group demonstrated a better ORR and PFS than the gefitinib group (ORR, 84% *v* 67% [$P < .001$]; PFS, 20.9 *v* 11.9 months; hazard ratio for death or disease progression, 0.490 [$P < .001$]), although PFS2 was not significantly different (20.9 *v* 18.0 months; $P = .092$). Median OS in the combination group was also significantly longer than in the gefitinib group (50.9 *v* 38.8 months; hazard ratio for death, 0.722; $P = .021$). The rate of grade ≥ 3 treatment-related adverse events, such as hematologic toxicities, in the combination group was higher than in the gefitinib group (65.3% *v* 31.0%); there were no differences in quality of life. One treatment-related death was observed in the combination group.

CONCLUSION Compared with gefitinib alone, gefitinib combined with carboplatin plus pemetrexed improved PFS in patients with untreated advanced NSCLC with EGFR mutations with an acceptable toxicity profile, although its OS benefit requires further validation.

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ASSOCIATED CONTENT

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Protocol

Author affiliations and support information (if applicable) appear at the end of this article.

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Written on behalf of the North-East Japan Study Group.

INTRODUCTION

Compelling evidence, including our NEJ002 study, demonstrates the superiority of first-line epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor (TKI) over platinum-based regimens in terms of progression-free survival (PFS) and quality of life (QOL) for patients with advanced non–small-cell lung cancer (NSCLC) harboring activating EGFR mutations.[1-10] However, overall survival (OS) was not improved in any of the these studies because of high crossover use of EGFR-TKIs. Moreover, in the NEJ002 study, approximately 30% of patients in the gefitinib group did not receive subsequent platinum doublet therapy, which suggests that the combined use of EGFR-TKI and platinum doublet might improve clinical outcomes. Although previous trials of EGFR-TKI combined with chemotherapy for nonselected patients with NSCLC failed to show clinical benefit, we hypothesized that EGFR-TKI combined with chemotherapy would be effective for the selected population that harbors mutated EGFR.

We conducted a randomized phase II study, NEJ005, that compared the concurrent use of gefitinib and carboplatin plus pemetrexed (GCP) with the sequential alternating use of the same agents in untreated patients with NSCLC with EGFR mutations. PFS was not different between the groups; however, median survival time (MST) was much longer in the concurrent group (41.9 *v* 30.7 months; hazard ratio [HR], 0.58; 95% CI, 0.34 to 0.97; $P = .036$), while toxicities were similar in each case.[11] Therefore, we selected the concurrent regimen for this phase III study to compare

its efficacy and safety with standard EGFR-TKI monotherapy as the first-line treatment of advanced NSCLC with EGFR mutations.

METHODS

Patient Population

This study was conducted according to the Declaration of Helsinki and was approved by the ethics review boards of each participating institution. Each patient provided written informed consent. The main eligibility criteria were chemotherapy naïve, stage IIIB or IV or relapsed nonsquamous NSCLC with EGFR mutations (exon 19 deletion, L858R, G719A, G719C, G719S, and L861Q), age 20 to 75 years, Eastern Cooperative Oncology Group performance status (PS) 0 to 1, and adequate organ function. The main exclusion criteria were serious concomitant systemic disorders, interstitial pneumonia, another primary malignancy, preexistence of T790M mutation, symptomatic brain metastases, and pregnancy.

Study Design and Treatment

Before randomization, patients were stratified according to sex, clinical stage (IIIB, IV, or postoperative relapse), type of EGFR mutation (exon 19 deletion, L858R, or other), and smoking history (yes or no). Eligible patients were randomly assigned to gefitinib (gefitinib 250 mg orally once every day) or GCP regimen (gefitinib 250 mg orally once every day combined with carboplatin area under the curve 5 and pemetrexed 500 mg/m^2 in a 3-week cycle for up to six cycles, followed by concurrent gefitinib and pemetrexed maintenance). In the GCP group, patients who were intolerant of gefitinib or chemotherapy were permitted to continue on the remaining agents. Although the protocol treatment should be terminated when progressive disease (PD) on the basis of RECIST version 1.1 was observed, treatment beyond progression was allowed at the discretion of investigators. In the gefitinib group, platinum-based regimens were recommended by the protocol as second-line chemotherapy.

Clinical Assessments

Radiographic assessments were performed at baseline and then every 2 months for the first year and every 3 months thereafter until PD. The investigators assessed response and PFS. The National Cancer Institute CTCAE (version 4.0) was used to grade adverse events. After PD, survival information, including postprotocol treatments, was collected every 6 months until death.

QOL Assessment

The European Organisation for Research and Treatment of Cancer (EORTC) QLQ-C30[12,13] was used to assess QOL. Patients completed the questionnaire before treatment, at 8 weeks, and then once every 6 months for the next 3 years. Within the EORTC QLQ-C30 is the Global health status/QOL scale (global QOL), which consists of items 29 and 30 and was of particular interest in this study. The global QOL score ranges from 2 points (worst) to 14 points (best).

Study End Points

This study tested multiple primary end points (PFS, PFS2, and OS), which were analyzed using a preplanned hierarchical sequential testing method, as explained in the Statistical Analysis section. The secondary end points included objective response rate, safety, and QOL.

PFS was evaluated for the period from the date of randomization until the date when PD or death as a result of any cause was first observed. PFS2 was defined as the period from the date of randomization until both platinum-based therapy and gefitinib became ineffective conceptually. Thus, the PFS2 in the gefitinib group was the period from randomization to PD of the second-line therapy or death, whichever occurred first, but it was the same as the PFS in the GCP group. After conducting this preplanned analysis, we noticed that the definition of our initial PFS2 was inappropriate as a comparator because only PFS2 in the gefitinib group included the time elapsed between first PD and initiation of the second-line treatment (so-called beyond-PD period). Therefore, we corrected the definition of PFS2 in the gefitinib group to subtract the beyond-PD period from the period from random assignment to PD of the second-line therapy or death. Although this modification was a posteriori, we performed a statistical analysis to ensure that comparisons that were based on the newly defined PFS2 were scientifically sound. We also performed an analysis that compared two groups by PFS2 with a same definition (events are true for second PD for both groups) as a reference. OS was evaluated for the period from the date of randomization to the date of death as a result of any cause.

The study was originally initiated with the single primary end point of OS. However, after the positive survival results from the AURA study[14] and several studies on immune checkpoint inhibitors, we amended the study protocol with regard to the primary end point, which was expected to minimize the risk of the OS end point being confounded by the increasing use of postprogression therapies. This protocol amendment was approved by the independent data and safety monitoring committee in February 2016.

Statistical Analysis

We sequentially analyzed the primary end points according to a preplanned hierarchical sequential testing method. First, the PFS of both groups was compared to verify superiority. We verified the superiority by comparing PFS2 only when the superiority was demonstrated in the first step. If superiority was demonstrated in both the first and the second steps, then superiority in OS was also tested. If superiority could not be verified in the first or second steps, then the trial was considered negative.

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The survival curves of PFS, PFS2, and OS were estimated using the Kaplan-Meier method and were compared between the GCP and gefitinib groups using a log-rank test. HRs and 95% CIs were calculated using Cox proportional hazards regression analysis. The response rate and rate of toxic effects were compared between groups using Fisher's exact test.

The sample size of this study was calculated on the basis of the following assumptions for OS: 4-year study period and a minimum follow-up of 3.5 years, MST of 27 months in the gefitinib group (referring to NEJ002), and HR of GCP versus gefitinib of 0.70. On the basis of these assumptions, a two-sided log-rank test with 80% power requires 168 patients in each arm to show a 5% significance. Therefore, total sample size was initially set at 340 patients. We assumed an HR of 0.7 for the PFS, PFS2, and OS end points and then confirmed that given this preplanned sample size, a sufficient statistical power was maintained for each. This meant that we did not have to change the target sample size.

RESULTS

Patient Characteristics

From October 2011 to September 2015, 345 patients were enrolled from 47 institutions in Japan and randomly assigned to two groups: 173 to the gefitinib group and 172 to the GCP group (Appendix Fig A1, online only). The characteristics of patients were well balanced between the groups (Table 1). Since sufficient numbers of events for PFS and PFS2 analyses were observed in October 2017, analysis for these primary end points was performed. The date of final data lock was September 3, 2018.

Efficacy

The objective response rate was 84% (95% CI, 79 to 90; 4% complete response and 79% partial response) in the GCP group and 67% (95% CI, 60 to 74; 3% complete response and 64% partial response) in the gefitinib group ($P < .001$). The waterfall plots of each group are shown in Appendix Figure A2 (online only). PFS was significantly longer in the GCP group than in the gefitinib group (median, 20.9 *v* 11.2 months; HR, 0.49; 95% CI, 0.39 to 0.62; $P < .001$; Fig 1A). The median duration of gefitinib treatment was longer in the GCP group (22.4 months; range, 0.5 to 59.8 months) than in the gefitinib group (11.6 months; range, 1.0 to 70.8 months). The median duration of pemetrexed maintenance therapy in the GCP group was 11.9 months (range, 0 to 57.9 months; Appendix Table A1, online only).

Although preplanned PFS2 analysis did not indicate any difference between the groups (20.9 *v* 20.7 months; HR, 0.99; $P = .90$; Appendix Fig A3, online only), modified analysis that subtracted the beyond-PD period in the gefitinib group revealed longer PFS2 in the GCP group (20.9 *v* 18.0 months; HR, 0.818; $P = .092$; Fig 1B).

TABLE 1. Patient Demographic and Clinical Characteristics at Baseline

Characteristic	Gefitinib, No. (%)	GCP, No. (%)
No. of patients	172	170
Sex		
Male	64 (37.2)	56 (32.9)
Female	108 (62.8)	114 (67.1)
Age, years		
Mean ± standard deviation	64.0 ± 8.4	64.8 ± 7.8
Range	37-75	34-75
Smoking status		
Never	97 (56.4)	96 (56.5)
Previous or current smoker	75 (43.6)	73 (42.9)
ECOG PS		
0	107 (62.2)	98 (57.6)
1	65 (37.8)	72 (42.4)
Histologic diagnosis		
Adenocarcinoma	170 (98.8)	168 (98.8)
Other	2 (1.2)	2 (1.2)
Clinical stage		
IIIA	1 (0.6)	0 (0.0)
IIIB	4 (2.3)	6 (3.5)
IV	137 (79.7)	139 (81.8)
Postoperative relapse	30 (17.4)	24 (14.7)
CNS metastasis		
Yes	38 (22.1)	50 (29.4)
Did not received irradiation	23 (13.4)	33 (19.4)
Pretreated with irradiation	15 (8.7)	17 (10.0)
No	134 (77.9)	120 (70.6)
Type of EGFR mutation		
Exon 19 deletion	95 (55.2)	93 (54.7)
L858R	67 (39.0)	69 (40.6)
Others	10 (5.8)	8 (4.7)

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; EGFR, epidermal growth factor receptor; GCP, gefitinib and carboplatin plus pemetrexed.

Furthermore, an additional analysis comparing the two groups by PFS2 with a same definition showed a large difference between the groups (32.5 *v* 20.7 months; HR, 0.59; $P < .001$; Fig 1C). Clinical status (PS, number of organs with metastasis, and percentage of patients with CNS metastasis) at first PD (PD1) was not different between the groups, although clinical status at second PD (PD2) in the gefitinib group, which indicated a failure of both EGFR-TKI and platinum doublet, was worse than that at PD1 in the GCP group (Appendix Table A2, online only), and the survival from PD2 of the gefitinib group was significantly worse than that from PD1 of the GCP group (Appendix Fig A4B, online only).

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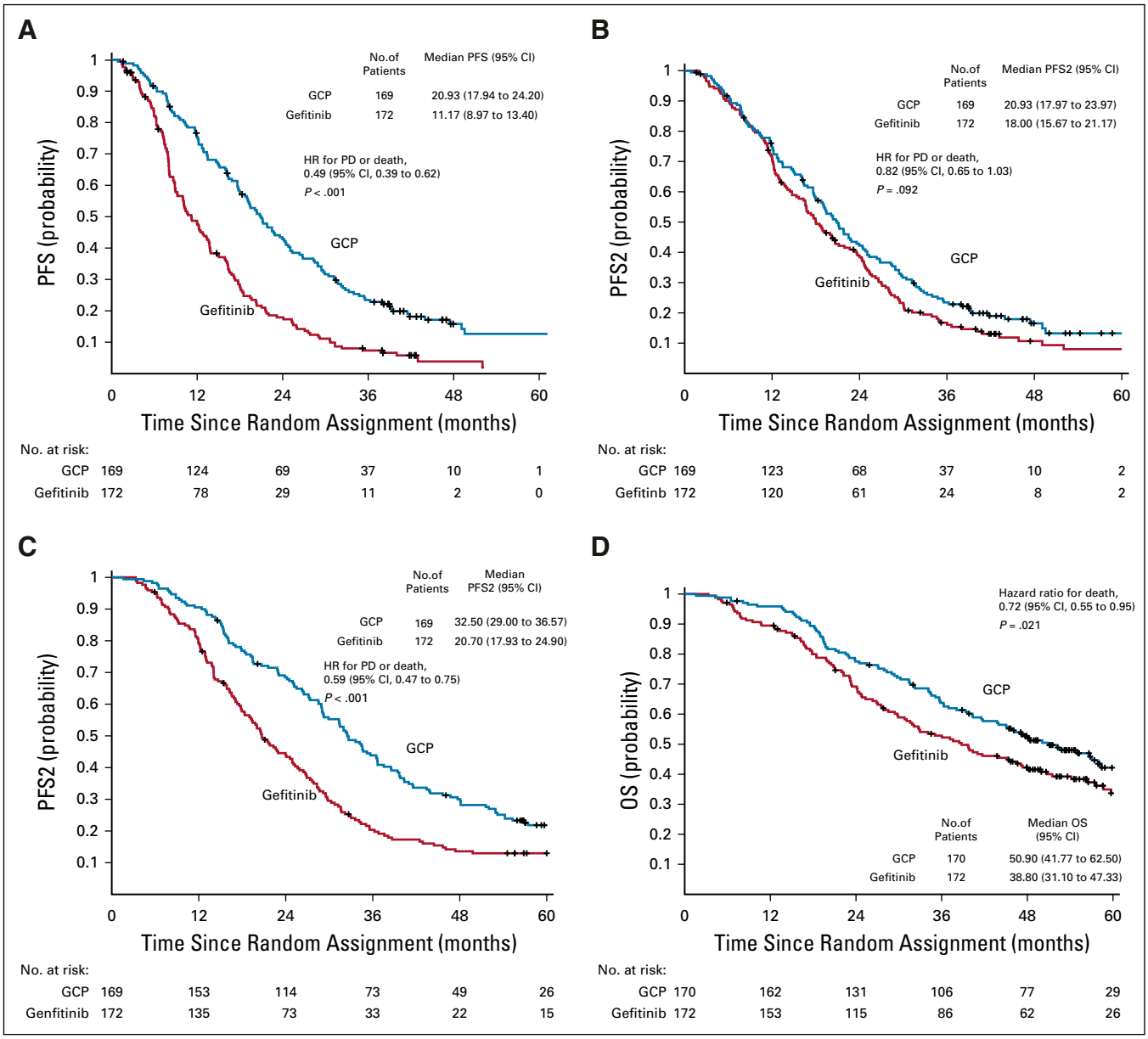

FIG 1. Progression-free survival (PFS) and overall survival (OS). Kaplan-Meier curves for (A) PFS, (B) PFS2, (C) PFS2 with a same definition, and (D) OS of patients treated with gefitinib and carboplatin plus pemetrexed (GCP) and those treated with gefitinib alone are shown. Bars indicate censored patients at the data cutoff point. HR, hazard ratio; PD, progressive disease.

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At the latest date cutoff point (with 195 events) when the median follow-up time was 45 months, the MSTs of the GCP and gefitinib groups were 50.9 and 38.8 months, respectively (HR, 0.722; 95% CI, 0.55 to 0.95, $P = .021$; Fig 1D). The forest plots of PFS and OS indicate the superiority of GCP over gefitinib, regardless of subgroup (Figs 2A and 2B). Details of postprotocol treatments in each group are listed in Appendix Table A3 (online only). According to the protocol recommendation, platinum-based chemotherapy was used for 77.4% of patients in the gefitinib group. Osimertinib was used for 37 patients (23.3%) in the gefitinib group and 29 (21.8%) in the GCP group. The MSTs of patients who received osimertinib were 74.4 months in the gefitinib group and not reached in the GCP group; these times are much longer than those observed in patients who did not receive osimertinib in each group (29.8 and 43.8 months, respectively; Appendix Fig A5, online only).

QOL

Of the 345 patients enrolled in this study, 279 (80.9%) completed the EORTC QLQ-C30 at baseline and more than once during the treatments. Demographics and disease characteristics were well balanced between the groups. Patients in either group had similar baseline global QOL scores (Fig 3). Although more decreased

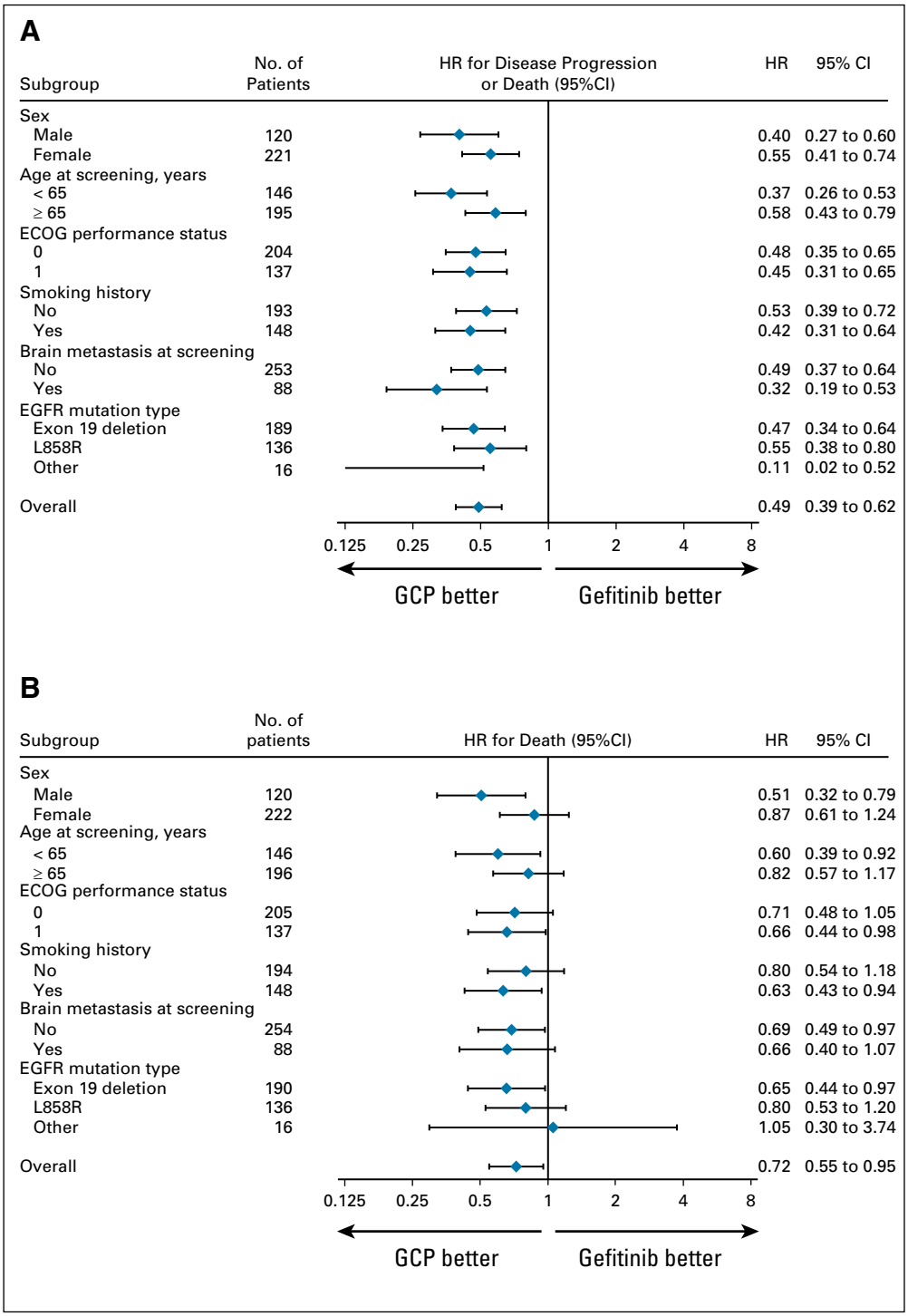

FIG 2. Subgroup analysis of progression-free survival (PFS) and overall survival (OS). Forest plots for (A) PFS and (B) OS are shown. A hazard ratio (HR) of < 1 implies a lower risk of disease progression or death with the gefitinib and carboplatin plus pemetrexed (GCP) regimen than with gefitinib alone. The Cox proportional hazards regression model includes randomly assigned treatment, the subgroup covariate of interest, and the treatment-by-subgroup interaction. EGFR, epidermal growth factor receptor.

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scores were observed in the GCP group at the 8-week point, no differences were observed between the groups at 6 months or later, indicating that the addition of pemetrexed to gefitinib did not impair global QOL. After 24 months, more patients in the GCP group (80 patients at 24 months, 75 at 30 months, and 68 at 36 months) completed the questionnaire than those in the gefitinib group (73 patients at 24 months, 58 at 30 months, and 56 at 36 months), which indicates that informative censoring occurred in the gefitinib group more because of poorer prognosis.

Safety

Adverse events related to the protocol treatment occurred in 98.2% of patients in the gefitinib group and 95.9% of patients in the GCP group. The rate of grade ≥ 3 treatment-related adverse events was 31.0% in the gefitinib group and 65.3% in the GCP group (Table 2; Appendix Table A4, online

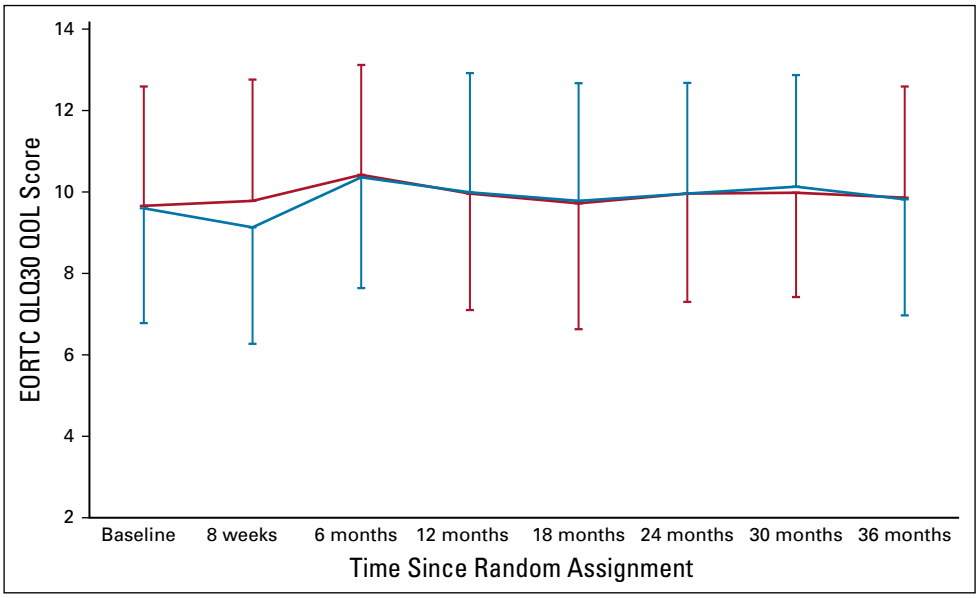

FIG 3. Global quality-of-life (QOL) scores. Global QOL scores assessed by European Organisation of Research and Treatment of Cancer (EORTC) QLQ-C30 in patients treated with gefitinib and carboplatin plus pemetrexed (red) and those treated with gefitinib alone (blue) are shown.

only). Neutropenia, anemia, and thrombocytopenia were more common in the GCP group (the rates of grade ≥ 3 for these toxicities were 31.2%, 21.2%, and 17.1% in the GCP group and 0.6%, 2.3%, and 0% in the gefitinib group, respectively). Meanwhile, liver toxicity was more frequent in the gefitinib group (the rate of grade ≥ 3 toxicity was 12.4% in the GCP group and 22.2% in the gefitinib group). A fatal adverse event as a result of severe infection occurred in one patient in the GCP group. The incidence of protocol treatment discontinuation because of adverse events was not different between the groups (gefitinib, 9.9%; GCP, 10.7%).

TABLE 2. Common Adverse Events

	Grade, No. (%)			
	Gefitinib (n = 171)		GCP (n = 170)	
Adverse Event	Any	≥ 3	Any	≥ 3
Treatment-related adverse events	168 (98.2)	53 (31.0)	163 (95.9)	111 (65.3)
Treatment-related adverse events with an incidence of > 10%				
Leukopenia	11 (6.4)	1 (0.6)	101 (59.4)	36 (21.2)
Neutropenia	7 (4.1)	1 (0.6)	101 (59.4)	53 (31.2)
Anemia	36 (21.2)	4 (2.3)	113 (65.5)	36 (21.2)
Thrombocytopenia	9 (5.3)	0 (0.0)	91 (53.5)	29 (17.1)
Liver dysfunction	102 (59.6)	38 (22.2)	103 (60.6)	21 (12.4)
Blood bilirubin increased	23 (13.5)	1 (0.6)	15 (8.8)	0 (0.0)
Creatinine elevation	11 (6.4)	0 (0.0)	43 (25.3)	0 (0.0)
Hyponatremia	6 (3.5)	1 (0.6)	34 (20.0)	5 (2.9)
Diarrhea	63 (36.8)	2 (1.2)	60 (35.3)	7 (4.1)
Vomiting	9 (5.3)	1 (0.6)	27 (15.9)	4 (2.4)
Stomatitis	29 (17)	0 (0.0)	52 (30.6)	1 (0.6)
Rash	138 (80.7)	5 (2.9)	110 (64.7)	7 (4.1)
Nail changes	57 (33.3)	2 (1.2)	44 (25.9)	5 (2.9)
Constipation	16 (9.4)	0 (0)	53 (31.2)	0 (0.0)
Anorexia	28 (16.4)	2 (1.2)	100 (58.8)	12 (7.1)
Edema limbs	7 (4.1)	0 (0.0)	29 (17.1)	3 (1.8)
Fatigue	20 (11.7)	0 (0.0)	67 (39.4)	7 (4.1)

Abbreviation: GCP, gefitinib and carboplatin plus pemetrexed.

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DISCUSSION

NEJ009 is the first phase III study in our knowledge to evaluate the efficacy and safety of the combination of EGFR-TKI and platinum doublet therapy compared with standard EGFR-TKI monotherapy for patients with advanced NSCLC harboring EGFR mutations. The GCP regimen consistently demonstrated promising results observed in our previous phase II trial.[11] Recently, another phase III study conducted in India also demonstrated the superior efficacy of this regimen compared with gefitinib alone (median PFS, 16 *v* 8 months; HR, 0.51).[15]

After approximately 10 to 15 months of initial response to EGFR-TKIs, almost all patients with EGFR-mutated NSCLC eventually develop resistance. In this study, the GCP regimen achieved much longer PFS than gefitinib alone, which means the combination regimen delayed the resistance. In addition, the GCP regimen might prevent the early resistance observed in several patients treated with EGFR-TKIs alone (Appendix Fig A2). The GCP regimen was also associated with a higher response rate than gefitinib alone. McCoach et al[16] reported that depth of response during first-line treatment with targeted therapy was associated with OS; the results of our trial are consistent with this finding. The survival after PD following first-line treatment was not different between the groups (Appendix Fig A4A).

A limitation of our study is that it failed to show the superiority of preplanned PFS2 analysis as a surrogate marker of OS. First, it did not consider the influence of the beyond-PD period (gefitinib: median, 1.0 month; range 0 to 34.5 months; GCP: median, 1.7 months; range, 0 to 40.4 months), which was included in the gefitinib group only. A previous study suggested that some patients would benefit from a beyond-PD use of EGFR-TKI, which is widely performed in general practice.[17] A corrected PFS2 analysis demonstrated that PFS2 was also superior in the GCP group. Second, although we were concerned about the large influence of the third-generation EGFR-TKI osimertinib on the OS of patients in this study, osimertinib was used for only 22% of patients after the first TKI treatment and did not influence the survival results, which suggests OS improvement with the GCP treatment. However, per protocol, OS should not undergo hierarchical testing; thus, the result was ad hoc and exploratory. The interpretation of the OS results needs to take into account the multiple comparisons issue.

Osimertinib was initially approved for T790M-positive NSCLC resistant to EGFR-TKI treatment on the basis of the AURA3 study.[14] The drug is also associated with significantly longer PFS than gefitinib or erlotinib when used as first-line treatment in patients with EGFR-mutated NSCLC.[18] The advantage of first-line osimertinib is its ability to extend PFS without a requirement for T790M genotyping, although guidelines recommend rebiopsy to examine T790M status in patients with EGFR-TKI resistance, which is not always possible in general practice. In fact, only 37% of patients resistant to gefitinib or erlotinib could cross over to osimertinib in the FLAURA study. Thus, we agree that patients with poor PS, including those with symptomatic CNS metastasis, should be treated with first-line osimertinib because they may have fewer opportunities to receive multiline treatments or additional examinations.

On the other hand, we believe that the GCP regimen is a strong candidate for extension of survival in patients in good condition. The OS data of first-line osimertinib are still emerging,[14] and the optimal treatment strategy after osimertinib has not been established. Since osimertinib is effective for EGFR-mutated NSCLC regardless of T790M existence, while gefitinib is only effective for those without T790M mutation, the first-line GCP treatment followed by osimertinib may be the best strategy. Ad hoc analysis in this study showed that the 3-year survival of patients treated with this sequence was 100%, although it had an exploratory nature that included some bias (eg, a T790M-positive tumor might be more indolent, so patients might have undergone rebiopsy at a higher rate; Appendix Fig A4). The risk of missing an opportunity of rebiopsy will decrease as liquid biopsy technology advances. We should therefore continue to improve treatment strategies on the basis of precision medicine and not on all-comer treatment for NSCLC with driver mutations.

With regard to safety, the GCP regimen led to an increase in neutropenia, anemia, and thrombocytopenia, although these toxicities were equivalent to or lower than those previously reported for carboplatin and pemetrexed.[19] Most toxicities were manageable, and the feasibility of the GCP regimen has been proven in this study. In fact, the discontinuation rate by study treatment was almost the same, and there was no difference in QOL between the groups. Thus, we believe that the safety margin of the GCP regimen is acceptable for patients with good PS.

Our study also suggests that a combination of TKI and cytotoxic chemotherapy could be successful in patients with other driver mutations, such as ALK or ROS1. With regard to immune therapy, the combination of immune checkpoint inhibitor and cytotoxic chemotherapy has already demonstrated excellent results.[20,21] Since TKIs cannot cure metastatic NSCLC, cytotoxic chemotherapy still plays an important role in anticancer treatment, and a strategy that uses up multiple effective agents should be investigated further. In this point of view, we are now conducting a randomized phase II study (UMIN identifier: UMIN000024438) that compares a combination of osimertinib and carboplatin plus pemetrexed with osimertinib alone for patients with EGFR-mutated, T790M-positive NSCLC as well as planning a study of the same combination as first-line treatment. Although the same combination

has been already investigated as a second-line therapy for T790M-positive NSCLC after treatment with first-line EGFR-TKI[22] and because the risk of thrombocytopenia and pneumonitis with osimertinib is much greater than that associated with gefitinib, we are concerned that the safety of the combination might be inferior to the current GCP regimen.

In conclusion, NEJ009 is the first phase III study to evaluate the efficacy of a combination of EGFR TKI and platinum doublet chemotherapy in patients with untreated advanced NSCLC with EGFR mutations. Although its OS benefit requires further validation, the GCP regimen improved PFS with an acceptable safety profile and a similar QOL profile compared with gefitinib alone.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST AND DATA AVAILABILITY STATEMENT

Disclosures provided by the authors and data availability statement (if applicable) are available with this article at DOI https://doi.org/10.1200/JCO.19.01488.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Gefitinib Alone Versus Gefitinib Plus Chemotherapy for Non–Small-Cell Lung Cancer With Mutated Epidermal Growth Factor Receptor: NEJ009 Study

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APPENDIX

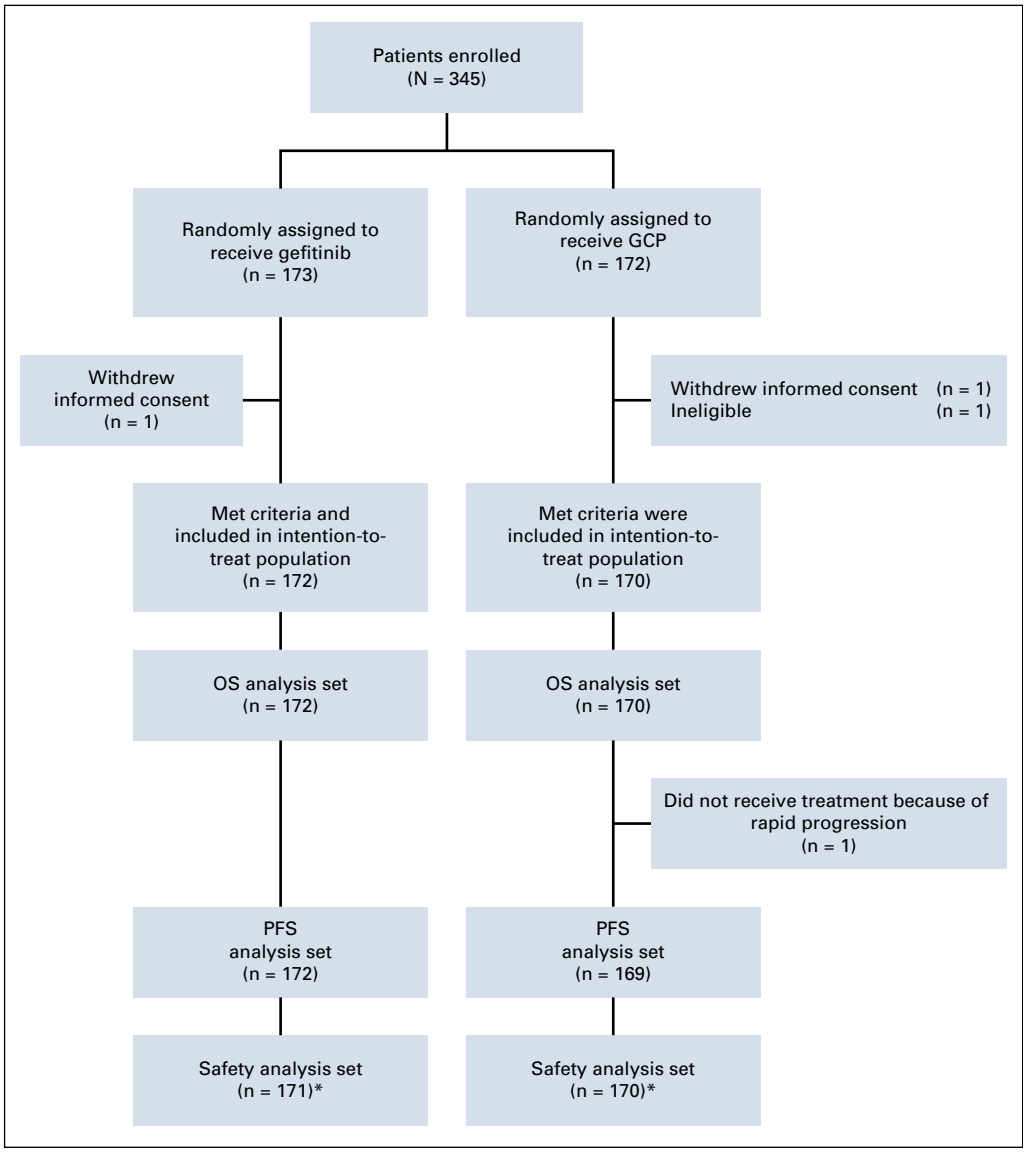

FIG A1. Flowchart of patient accumulation. All patients except one ineligible patient and two patients who withdrew consent were randomly assigned to the gefitinib group or gefitinib and carboplatin plus pemetrexed (GCP) group. One patient in the gefitinib group did not receive gefitinib alone but was treated with the GCP regimen at the patient's request; thus, this patient's safety data were evaluated as if the patient was in the GCP group, whereas the progression-free survival (PFS) and overall survival (OS) data were evaluated as if the patient was in the gefitinib group based on the intention to treat. One patient in the GCP group was excluded from the PFS analysis because the patient did not receive any protocol treatment; this patient was included in the OS analysis only. (*) One patient received GCP instead of gefitinib monotherapy with a breach of allocation.

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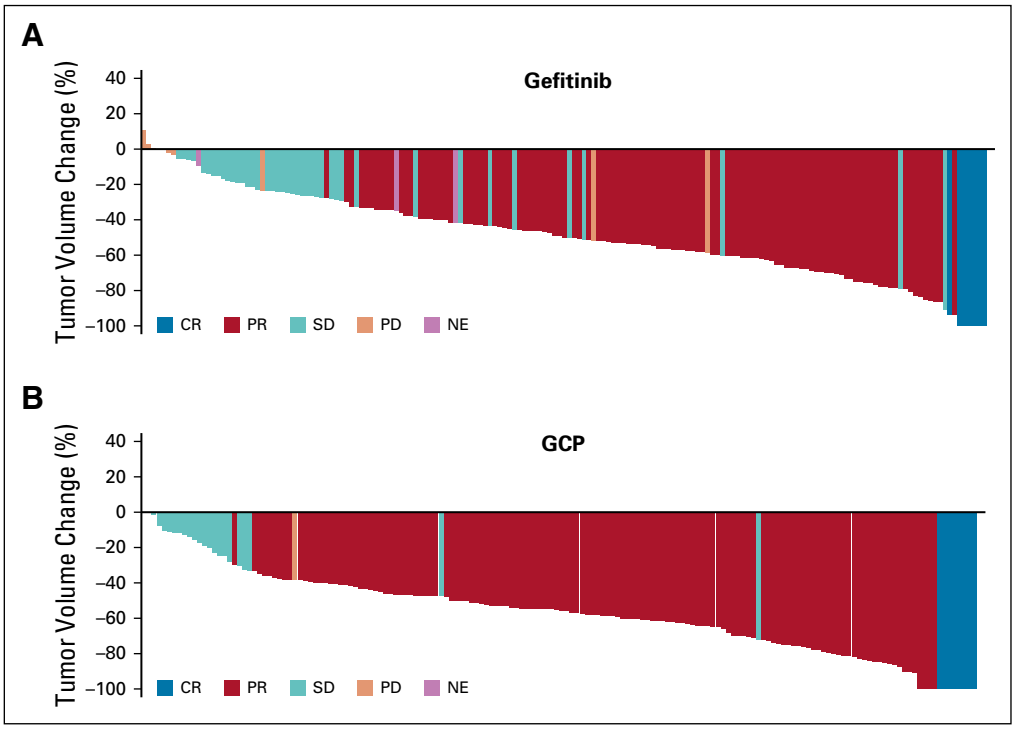

FIG A2. Best percent change from baseline in target lesions. Waterfall plots for best percent change in target lesion size are shown for (A) patients who received gefitinib alone versus (B) patients who received gefitinib and carboplatin plus pemetrexed (GCP). CR, complete response; NE, not estimable; PD, progressive disease; PR, partial response; SD, stable disease.

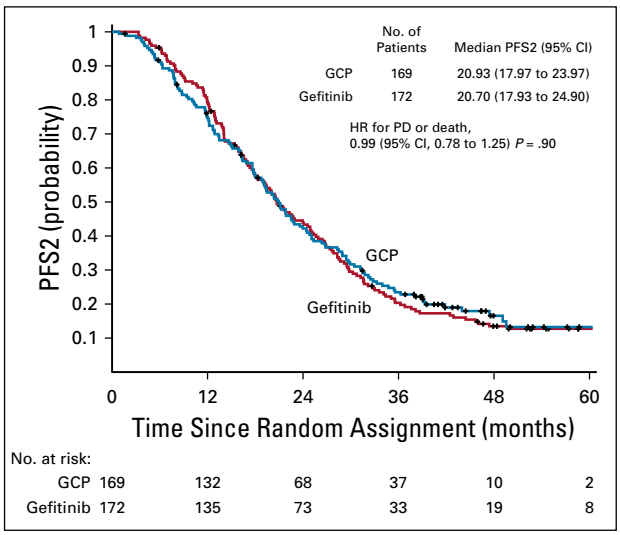

FIG A3. Preplanned progression-free survival 2 (PFS2) comparison. Kaplan-Meier curves for PFS2 of patients treated with gefitinib and carboplatin plus pemetrexed (GCP) and those treated with gefitinib alone are shown. HR, hazard ratio; PD, disease progression.

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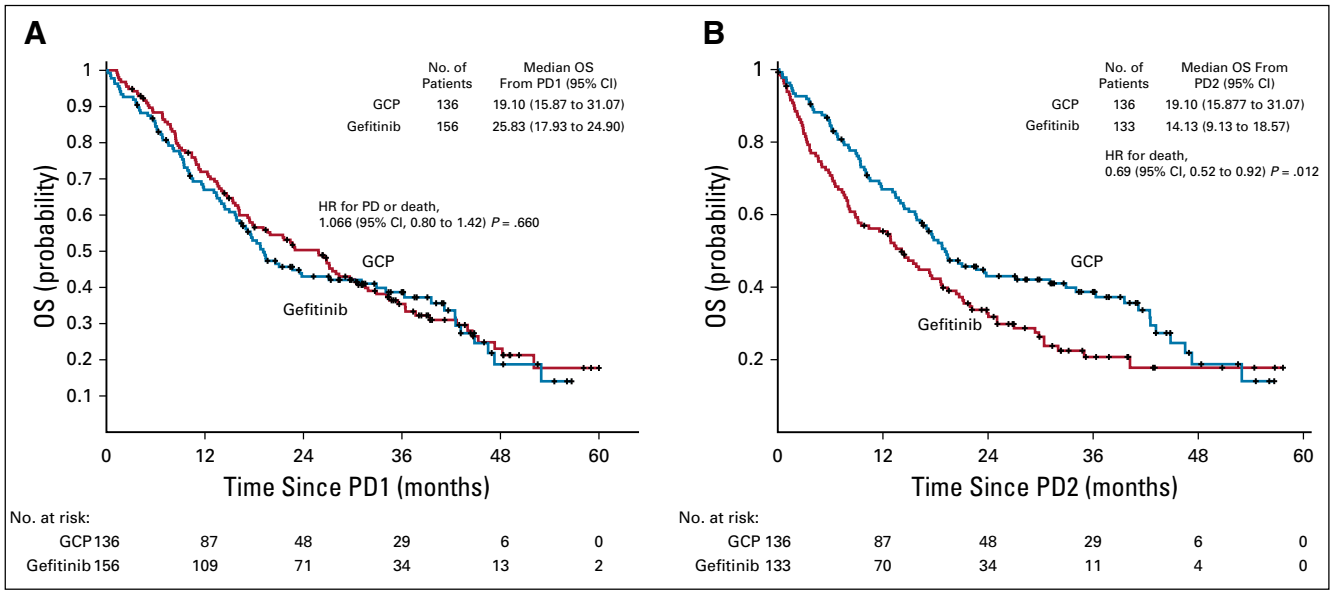

FIG A4. Overall survival (OS) from the first and the second disease progression. (A) Kaplan-Meier curves for OS from disease progression in the first-line treatment (PD1) of patients treated with gefitinib and carboplatin plus pemetrexed (GCP) and those treated with gefitinib alone are shown. (B) OS is shown from PD1 of patients treated with GCP and that from the second PD (PD2) of patients treated with gefitinib alone. Bars indicate censored patients at the data cutoff point. HR, hazard ratio.

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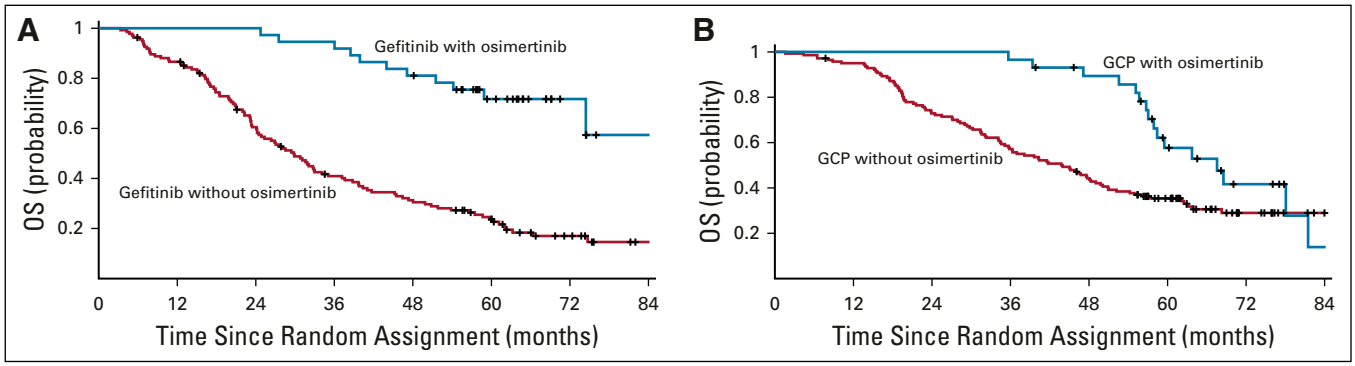

FIG A5. Overall survival (OS) of patients treated with or without osimertinib. Kaplan-Meier curves for (A) OS of patients treated with or without osimertinib in the gefitinib group and (B) patients in the gefitinib and carboplatin plus pemetrexed (GCP) group are shown. Bars indicate censored patients at the data cutoff point.

TABLE A1. Exposure to Study Treatment

Variable	Gefitinib (n = 172)	GCP (n = 169)
Duration of gefitinib treatment, months		
Median (range)	11.6 (1.0-70.8)	22.4 (0.5-59.8)
Carboplatin plus pemetrexed, No. (%)		168 (99.4)
Median cycles (range)	—	4 (1-6)
Median duration, months (range)	—	2.6 (0-7.7)
Pemetrexed maintenance, No. (%)		135 (79.9)
Median cycles (range)	—	16 (1-75)
Median duration, months (range)	—	11.9 (0-57.9)

Abbreviation: GCP, gefitinib and carboplatin plus pemetrexed.

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TABLE A2. Clinical Status at First and Second Disease Progression

Disease Progression	Gefitinib (n = 172), No. (%)	GCP (n = 169), No. (%)
First disease progression	153	135
ECOG PS		
0-1	134 (87.6)	116 (85.9)
2	8 (5.2)	12 (8.9)
3-4	3 (2.0)	4 (2.9)
Median organs with metastasis (range)	1 (0-5)	1 (0-7)
Brain metastasis	38 (24.8)	48 (35.6)
Second disease progression	128	—
ECOG PS		—
0-1	88 (68.8)	
2	19 (14.8)	
3-4	11 (8.6)	
Median organs with metastasis (range)	2 (0-6)	—
Brain metastasis	38 (29.7)	—

Abbreviation: ECOG PS, Eastern Cooperative Oncology Group performance status; GCP, gefitinib and carboplatin plus pemetrexed.

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TABLE A3. Post–First-Line Treatments

Variable	Gefitinib (n = 172), No. (%)	GCP (n = 170), No. (%)
No. with ongoing first-line treatment	13	37
No. who discontinued first-line treatment	159	133
No treatment	8 (5.0)	17 (12.8)
Any treatments	151 (95.0)	116 (87.2)
Platinum doublet with or without bevacizumab	123 (77.4)	26 (19.5)
Pemetrexed	33 (20.8)	6 (4.5)
Docetaxel with or without ramucirumab	49 (30.8)	57 (42.9)
Tegafur, gimeracil, and oteracil	21 (13.2)	19 (14.3)
Gefitinib or erlotinib	66 (41.5)	48 (36.1)
Afatinib	38 (23.9)	42 (31.6)
Osimertinib	37 (23.3)	29 (21.8)
Immune checkpoint inhibitors	20 (12.6)	17 (12.8)
Other	34 (21.4)	21 (15.8)

Abbreviation: GCP, gefitinib and carboplatin plus pemetrexed.

TABLE A4. Adverse Events

	Grade, No. (%)			
	Gefitinib (n = 171)		GCP (n = 170)	
Adverse Event	**Any**	**≥ 3**	**Any**	**≥ 3**
Treatment-related adverse events	168 (98.2)	53 (31.0)	163 (95.9)	111 (65.3)
Infection	6 (3.5)	0 (0.0)	22 (12.9)	9 (5.3)
Febrile neutropenia	0 (0.0)	0 (0.0)	5 (2.9)	5 (2.9)
Constipation	16 (9.4)	0 (0.0)	53 (31.2)	0 (0.0)
Diarrhea	63 (36.8)	2 (1.2)	60 (35.3)	7 (4.1)
Stomatitis	29 (17.0)	0 (0.0)	52 (30.6)	1 (0.6)
Vomiting	9 (5.3)	1 (0.6)	27 (15.9)	4 (2.4)
Anorexia	28 (16.4)	2 (1.2)	100 (58.8)	12 (7.1)
Fatigue	20 (11.7)	0 (0.0)	67 (39.4)	7 (4.1)
Rash	138 (80.7)	5 (2.9)	110 (64.7)	7 (4.1)
Alopecia	9 (5.3)	0 (0.0)	17 (10.0)	0 (0.0)
Nail changes	57 (33.3)	2 (1.2)	44 (25.9)	5 (2.9)
Edema limbs	7 (4.1)	0 (0.0)	29 (17.1)	3 (1.8)
Pneumonitis	6 (3.5)	1 (0.6)	11 (6.5)	4 (2.4)
Dyspnea	12 (7.0)	0 (0.0)	15 (8.8)	2 (1.2)
Motor neuropathy	3 (1.8)	0 (0.0)	1 (0.6)	0 (0.0)
Sensory neuropathy	4 (2.3)	0 (0.0)	14 (8.2)	1 (0.6)
Allergic reaction	0 (0.0)	0 (0.0)	6 (3.5)	2 (1.2)
Arthralgia/myalgia	7 (4.1)	0 (0.0)	6 (3.5)	0 (0.0)
Leukopenia	11 (6.4)	1 (0.6)	101 (59.4)	36 (21.2)
Neutropenia	7 (4.1)	1 (0.6)	101 (59.4)	53 (31.2)
Anemia	36 (21.1)	4 (2.3)	113 (66.5)	36 (21.2)
Thrombocytopenia	9 (5.3)	0 (0.0)	91 (53.5)	29 (17.1)
Blood bilirubin increased	23 (13.5)	1 (0.6)	15 (8.8)	0 (0.0)
Liver dysfunction	102 (59.6)	38 (22.2)	103 (60.6)	21 (12.4)
Creatinine elevation	11 (6.4)	0 (0.0)	43 (25.3)	0 (0.0)
Hyperkalemia	10 (5.8)	1 (0.6)	13 (7.6)	0 (0.0)
Hyponatremia	6 (3.5)	1 (0.6)	34 (20.0)	5 (2.9)
Hypercalcemia	0 (0.0)	0 (0.0)	3 (1.8)	0 (0.0)
Hypoalbuminemia	6 (3.5)	0 (0.0)	14 (8.2)	0 (0.0)
Dysgeusia	8 (4.7)	0 (0.0)	12 (7.1)	0 (0.0)
Hypertension	10 (5.8)	3 (1.8)	9 (5.3)	2 (1.2)
Pain	8 (4.7)	0 (0.0)	10 (5.9)	0 (0.0)
Dry skin	10 (5.8)	0 (0.0)	6 (3.5)	0 (0.0)
Alkaline phosphatase increased	7 (4.1)	0 (0.0)	8 (4.7)	0 (0.0)
Lactate dehydrogenase increased	8 (4.7)	0 (0.0)	7 (4.1)	0 (0.0)
Hypocalcemia	7 (4.1)	0 (0.0)	8 (4.7)	0 (0.0)
Nausea	6 (3.5)	0 (0.0)	9 (5.3)	1 (0.6)
Hypoproteinemia	7 (4.1)	0 (0.0)	7 (4.1)	0 (0.0)
Blood chlorine increased	6 (3.5)	0 (0.0)	7 (4.1)	0 (0.0)
Hyperglycemia	7 (4.1)	0 (0.0)	6 (3.5)	0 (0.0)

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TABLE A4. Adverse Events (continued)

	Grade, No. (%)			
	Gefitinib (n = 171)		GCP (n = 170)	
Adverse Event	**Any**	**≥ 3**	**Any**	**≥ 3**
Skin hyperpigmentation	2 (1.2)	0 (0.0)	10 (5.9)	0 (0.0)
Lymphocyte count decreased	5 (2.9)	0 (0.0)	6 (3.5)	2 (1.2)
Blood urea nitrogen increased	4 (2.3)	0 (0.0)	6 (3.5)	0 (0.0)
Hypertriglyceridemia	3 (1.8)	0 (0.0)	7 (4.1)	0 (0.0)
Hemorrhage	8 (4.7)	1 (0.6)	2 (1.2)	0 (0.0)
Pruritus	6 (3.5)	0 (0.0)	5 (2.9)	0 (0.0)
Dizziness	5 (2.9)	0 (0.0)	4 (2.4)	0 (0.0)
Hypokalemia	5 (2.9)	4 (2.3)	3 (1.8)	0 (0.0)
Gastritis	6 (3.5)	0 (0.0)	1 (0.6)	0 (0.0)
Cough	5 (2.9)	0 (0.0)	2 (1.2)	0 (0.0)
CPK increased	3 (1.8)	0 (0.0)	3 (1.8)	0 (0.0)
Weight loss	2 (1.2)	0 (0.0)	4 (2.4)	2 (1.2)
Proteinuria	2 (1.2)	0 (0.0)	3 (1.8)	0 (0.0)
Insomnia	2 (1.2)	0 (0.0)	3 (1.8)	0 (0.0)
Serum amylase increased	3 (1.8)	2 (1.2)	1 (0.6)	1 (0.6)
Conjunctivitis	1 (0.6)	0 (0.0)	3 (1.8)	0 (0.0)
GI hemorrhage	1 (0.6)	0 (0.0)	3 (1.8)	3 (1.8)
Arrhythmia	2 (1.2)	0 (0.0)	2 (1.2)	0 (0.0)
Abdominal distension	1 (0.6)	0 (0.0)	3 (1.8)	0 (0.0)
Blood chlorine decreased	2 (1.2)	0 (0.0)	1 (0.6)	0 (0.0)
Cholesterol high	1 (0.6)	0 (0.0)	2 (1.2)	0 (0.0)
Hyperuricemia	1 (0.6)	0 (0.0)	2 (1.2)	0 (0.0)
Esophagitis	2 (1.2)	0 (0.0)	1 (0.6)	1 (0.6)
Fever	1 (0.6)	0 (0.0)	2 (1.2)	1 (0.6)
Anxiety	2 (1.2)	0 (0.0)	1 (0.6)	1 (0.6)
Delirium	0 (0.0)	0 (0.0)	2 (1.2)	1 (0.6)
Ileus	0 (0.0)	0 (0.0)	2 (1.2)	2 (1.2)
Depressed level of consciousness	1 (0.6)	1 (0.6)	1 (0.6)	0 (0.0)
Gastric ulcer	2 (1.2)	1 (0.6)	0 (0.0)	0 (0.0)
Vertigo	2 (1.2)	0 (0.0)	0 (0.0)	0 (0.0)
Eye discharge	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Hiccups	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Periodontal disease	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Tinnitus	1 (0.6)	0 (0.0)	1 (0.6)	0 (0.0)
Hand-foot syndrome	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Fluid retention	0 (0.0)	0 (0.0)	2 (1.2)	1 (0.6)
Shingles	2 (1.2)	0 (0.0)	0 (0.0)	0 (0.0)
Fall	1 (0.6)	0 (0.0)	1 (0.6)	0 (0.0)
Rhinitis	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Edema face	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)
Blurred vision	0 (0.0)	0 (0.0)	2 (1.2)	0 (0.0)

(continued on following page)

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TABLE A4. Adverse Events (continued)

	Grade, No. (%)			
	Gefitinib (n = 171)		GCP (n = 170)	
Adverse Event	**Any**	**≥ 3**	**Any**	**≥ 3**
Activated partial thromboplastin time prolonged	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
BNP increased	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Depression	0 (0.0)	0 (0.0)	1 (0.6)	1 (0.6)
Lower-limb phlebitis	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Corneal ulcer	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Dry eye	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Endophthalmitis	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Vasovagal reaction	0 (0.0)	0 (0.0)	1 (0.6)	1 (0.6)
Thromboembolic event	0 (0.0)	0 (0.0)	1 (0.6)	1 (0.6)
Angular cheilitis	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Herpes labialis	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Dry mouth	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Dysarthria	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Hyperproteinemia	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Fracture	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Hemorrhoids	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Neoplasms	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Meningismus	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Glossitis	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Dehydration	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Cholangitis	0 (0.0)	0 (0.0)	1 (0.6)	1 (0.6)
Headache	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Urinary incontinence	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Nasal discharge	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Urinary frequency	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Radiation recall reaction (dermatologic)	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)
Brachial plexopathy	1 (0.6)	1 (0.6)	0 (0.0)	0 (0.0)
Hoarseness	1 (0.6)	0 (0.0)	0 (0.0)	0 (0.0)
Otitis media	0 (0.0)	0 (0.0)	1 (0.6)	0 (0.0)

Abbreviations: BNP, brain natriuretic peptide; CPK, creatine phosphokinase; GCP, gefitinib and carboplatin plus pemetrexed.

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